Pritish Chakraborty

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Education

- **Indian Institute of Technology, Bombay** — Mumbai, Maharashtra
  MS + PhD (dual degree) in Computer Science; GPA: 9.14/10 — *July 2021 – June 2024 — July 2024 - present*
- **Manav Rachna International University** — Delhi NCR
  BTech in Computer Science and Engineering; GPA: 8.27/10 — *July 2010 – May 2014*

Research Interests

Information Retrieval for Structured Objects, Graph Retrieval and interfacing with LLMs, ML on Graphs, Neural Tools for Combinatorial Algorithms, Geometric Deep Learning, Generative Models, Applications (AI4Science)

Relevant Research Work

- **Position: Neural Approximation Is Rarely Justified for Hard Combinatorial Problems**
  Advisors: Prof. Abir De and Prof. Soumen Chakrabarti — *November 2025 - January 2026.*
  Developed an accounting framework for neural approaches to NP-hard combinatorial problems, arguing that neural surrogates should not be treated as wholesale replacements for classical solvers unless required by end-to-end differentiability, system-level serving constraints such as vector indexing, or solver augmentation. Includes analysis of training and label-generation costs, distribution shift, lack of feasibility/approximation guarantees, and over-dependence on classical combinatorial decoders. Also proposed a practical diagnostic checklist for deciding when neural methods provide genuine value over established combinatorial algorithms. **Paper accepted to ICML 2026**. (Please see: https://pritishc.com/posts/2026/06/neural-combinatorial-blog/)
- **Multi-Way Representation Alignment**
  Mentor: Donato Crisostomi (LogML) — *August 2025 - January 2026.*
  Contributed to the development of a multi-way representation alignment framework that maps independently trained models into a shared latent universe, avoiding the quadratic cost and inconsistency of pairwise alignment. The proposed Geometry-Corrected Procrustes Alignment combines an orthogonal Procrustes-based universe with a lightweight shared correction for improved any-to-any retrieval while preserving a reusable coordinate system. Designed and executed the cross-lingual retrieval and universal clustering experiments, and was tasked with running and analyzing all experiments validating the method across representation alignment settings. **Paper accepted to ICML 2026**. (Please see: https://pritishc.com/posts/2026/06/multiway-alignment-blog/)
- **A Dense Subset Index for Collective Query Coverage (DISCo)**
  Advisors: Prof. Abir De and Prof. Soumen Chakrabarti — *June 2025 - October 2025.*
  Developed dense retrieval framework for **collective query coverage** that retrieves a set of documents jointly, so that different aspects of a query are covered, as opposed to independent ranking. Includes contributions to the core indexing and retrieval mechanism, various definitive baselines and ablation experiments, and performance benchmarking of the end-to-end system. **Paper accepted to ICLR 2026**. (Please see: https://tinyurl.com/disco-paper)
- **Contextual Tokenization for Graph Inverted Indices (CoRGII)**
  Advisors: Prof. Abir De and Prof. Soumen Chakrabarti — *February 2025 - May 2025*
  Developed **efficient, differentiable sparse retrieval pipeline** for the purpose of graph retrieval. Consists of neural tokenizer for discretized latent vocabulary, impact scorer network for differentiable analog of TF-IDF scoring, and multi-probe expansion using GPU-bound inverted index for greater recall. Leverages contextual dense graph representations to generate discrete codes, which support highly optimized inverted indices. **Paper accepted to NeurIPS 2025, Learning on Graphs 2025**. (Please see: https://pritishc.com/posts/2025/10/corgii-blog/)
- **Differentiable Adversarial Attacks against Marked Temporal Point Processes (PermTPP)**
  Advisors: Prof. Abir De and Prof. Srikanta Bedathur — *Nov 2023 - June 2024*
  Developed new class of **fully differentiable** adversarial attacks for marked temporal point process (MTPP) models, while outlining why standard attacks (PGD, etc) are not suitable. The attack corrupts input continuous-time event sequence data in a way that perturbs order relationship between elements of a sequence. Subsequently, developed algorithm to train MTPP models to become robust to this class of attacks. Evaluated the method on four real-world datasets, in addition to datasets from specific domains. Results show that our method out-performs standard attacks. **Paper accepted to AAAI 2025, ACM TIST 2026**. (Please see: https://tinyurl.com/mtpp-adv)

- **Learning and Maximizing Influence in Social Networks Under Capacity Constraints**
  Advisors: Prof. Abir De and Prof. Sayan Ranu *Jan 2022 - Aug 2022.*
  Developed and implemented an MC simulation based method to obtain an (approximately) optimal solution to the **top-K influence maximization problem**. Helped develop **alpha-submodular** theoretical guarantees in the process. Developed **differentiable deep submodular neural network** to predict the top-K maximized influential nodes in a graph. Both simulator and ML models outperformed state-of-the-art influence maximization methods. **Paper accepted to WSDM 2023**. (Please see: https://tinyurl.com/topk-im)

Work Experience

- **Google Research India** Hybrid
  Student Researcher *July 2023 - November 2023*
  - **Responsibilities** End-to-end research and execution. Building Google-specific tools based on my work on influence maximization.
  - **Projects** Research on the robust influence maximization problem. Development of general foundation models for information diffusion. Development of fast neural approximators for replacement of MC simulations.
- **Instahyre** New Delhi, India
  Software Engineering/Head of Engineering *July 2015 - May 2019*
  - **Responsibilities** Handled the complete tech stack and deployment pipeline for multiple feature release cycles. Mentored and brought junior engineers upto speed.
  - **Research and Execution** Undertook systems research for various aspects of the codebase, both during maintenance and when extending features from scratch. Aligned this research with business agenda - for example, deciding to scale parts of the system, vertically or horizontally, depending on projected user count and user load across time. Additionally, undertook user experience research when deciding to release certain big-ticket features, for example performing large scale A/B tests.
  - **Projects** Worked on several important features - the matching engine, the Elasticsearch-powered search platform, internal tools - end to end. Took responsibility for a streamlined UX and production management.

Extra-Curriculars

- **Reviewer**: AAAI 2023, ICLR 2026, NeurIPS 2026.
- **Fellowships**: Microsoft Research India PhD Award, 2026, Qualcomm Innovation Fellowship, 2025.
- **Grants**: ACM IARCS Research Travel Grant, 2025, NeurIPS 2025 Scholar Award.
- **Workshops**: DiffCoALG Workshop @ NeurIPS 2025 (https://sites.google.com/view/diffcoalg2025) as Web Chair and volunteer.
- **Summer schools**: Machine Learning Summer School, 2026 (Columbia University, NY, mlss.cc), London Geometry and Machine Learning Summer School, 2025 (logml.ai). Our group won the **prize for best project** at LogML.

Technical Skills

- **Languages**: Python, Javascript, C/C++, SQL, other languages such as C# and Java
- **Technologies**: Pytorch, Pytorch Geometric, Django, Angular.js, React, REST, other web technologies

Courses Taken

Foundations of Machine Learning, Algorithms and Complexity, Retrieval on Text and Graphs, Advanced Machine Learning, Optimization in Machine Learning, Organization of Web Information (NLP+IR), Real Analysis, Basic Algebra